


# : An Overview

# Outline

- Introduction and definition cloud computing
- Characteristics of Cloud Computing
- Service delivery  models of cloud computing
- Deployment models of cloud computing
- Cloud Role and Boundaries
- Technologies in Cloud Computing
- Key drivers to adopt the cloud
- Edge and fog computing



# Introduction

- In 1965, **Corbato** *et al*. provide a business model for on demand computing resources that is based on *pay-as you-go* manner - means customers get to choose whatever resources (like- CPUs, memory, bandwidth, platforms, etc) from the vendor.
- Cloud computing developed in 2005/2006 as a new distributed computing model with the aimed of attaining the computing as a utility like public user obtain services from traditional public utility services such as water, electricity, gas and telephone.



# *Introduction…*

- Cloud computing is taking services and moving them outside an organization's firewall. Applications, storage and other services are accessed via the Web.

- The provision of dynamically scalable and often virtualized resources as a service over the Internet on a utility basis.
- Users need not have knowledge of, expertise in, or control over the technology infrastructure in the 'cloud' that supports them."



# *Cloud Definition*

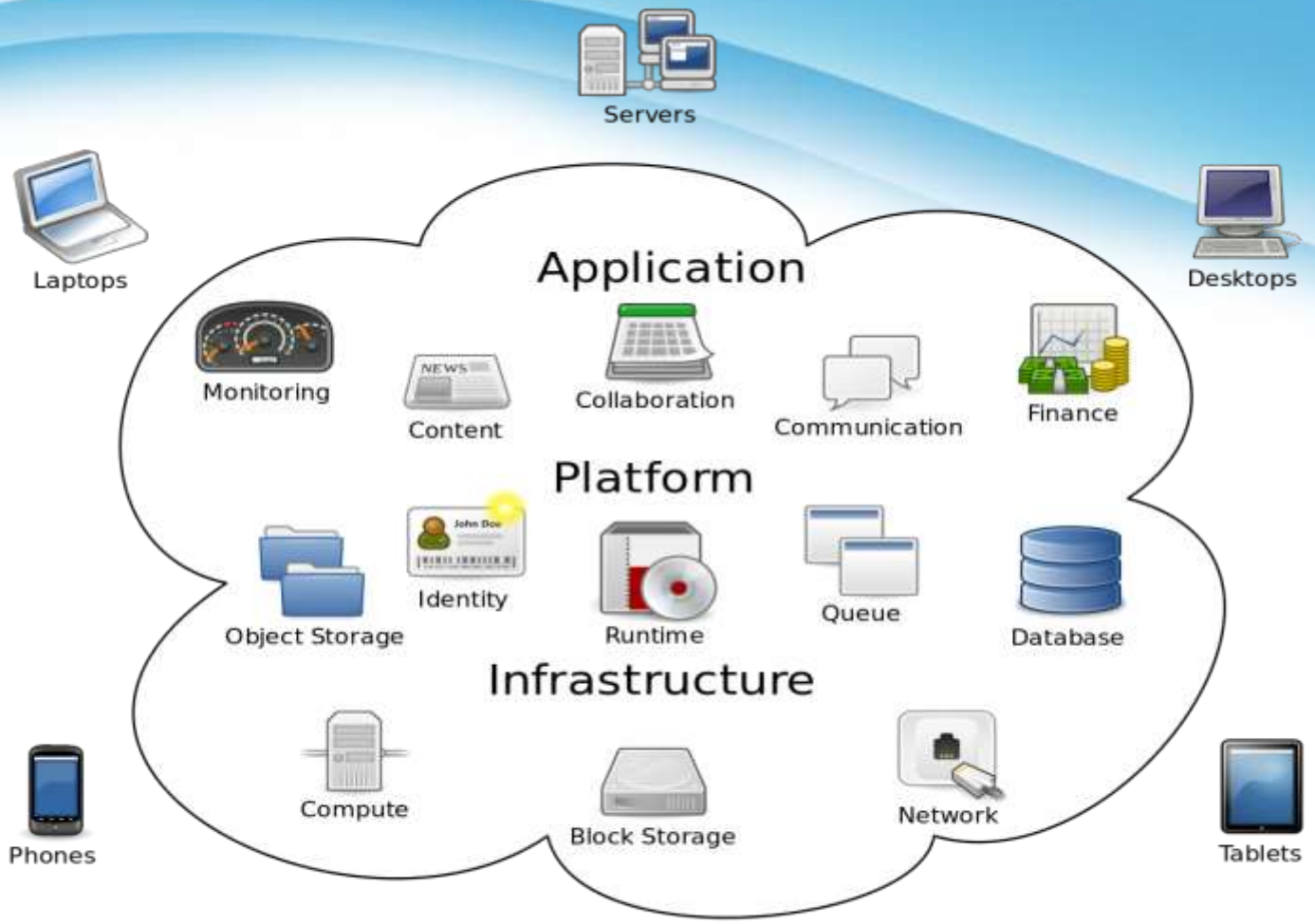
The **NIST** has a more comprehensive definition of cloud computing as-

"*A model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g.,* ***networks, servers, storage, applications, and services****) that can be rapidly provisioned and released with minimal management effort or service provider interaction*."

# *Cloud Definition*

- Cloud computing is the delivery of computing services—including servers, storage, databases, networking, software, analytics, and intelligence— over the internet ("the cloud") to offer faster innovation, flexible resources, and economies of scale.
- By providing computing resources on demand, cloud computing avoids the need to install of physical servers, run own software, and manage the own databases.

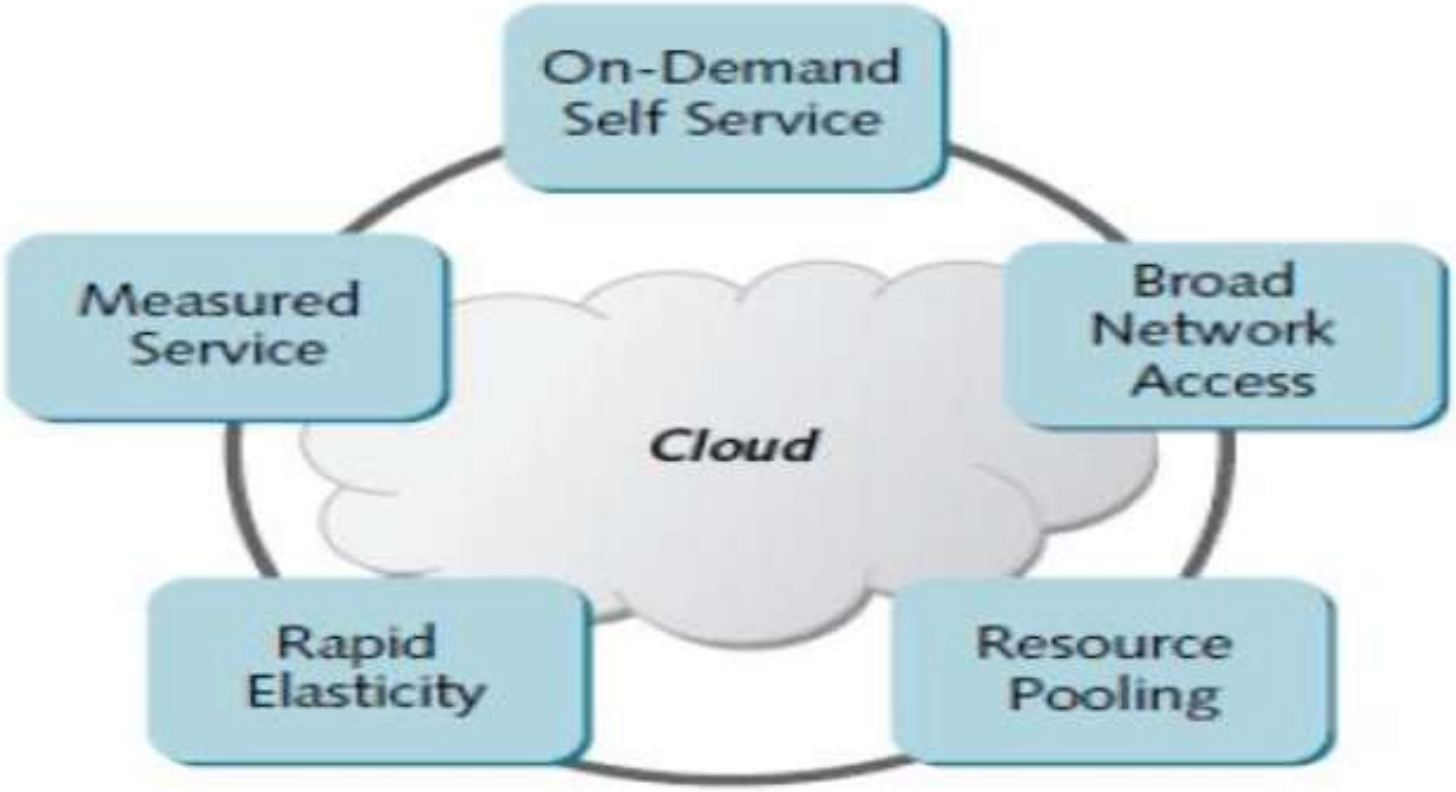
- By making the use of physical hardware more efficient and allowing one physical machine to serve many different needs and organizations, cloud computing reduces the cost of managing and accessing computing resources.




# Cloud computing

# Characteristics of cloud computing

# Characteristics of cloud computing



# On-demand self-service

- It enables consumers to directly request, manage and access the services through the web services and management interfaces without any human interaction.
- Computing resources are instantly available to users as per their requests.



# Broad network access

- The data and services are presented in the cloud are accessible through using any standard device like mobile phones, PC, desktop, laptop.
- Cloud-based services  are accessed from any location any time provided that there is adequate IP networking.



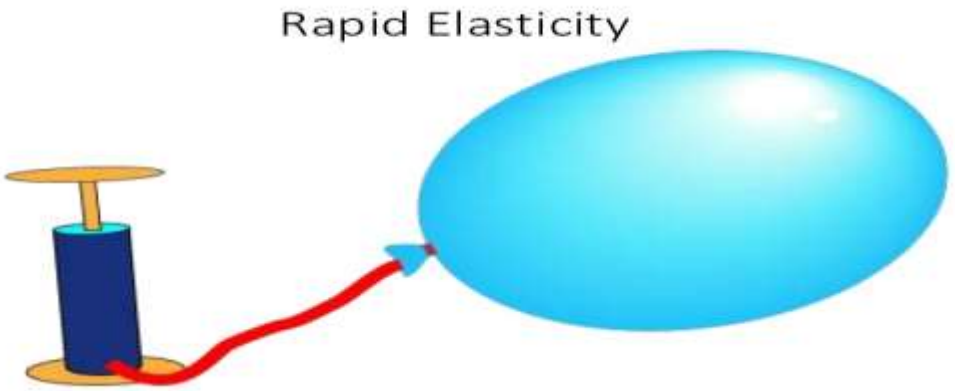
# Resource pooling

- The cloud provider provides a large physical or virtual computing resources that are shared among multiple users.
- These resources are allocated dynamically in a multi-tenant environment.



# Rapid elasticity

- Resources are  scaled according to the consumer requirements. Customers have unlimited resources that can be purchased as needed in pay-per use manner.
- Elastically increases service capacity during busy periods, and  reduces capacity during customers' off-peak periods.

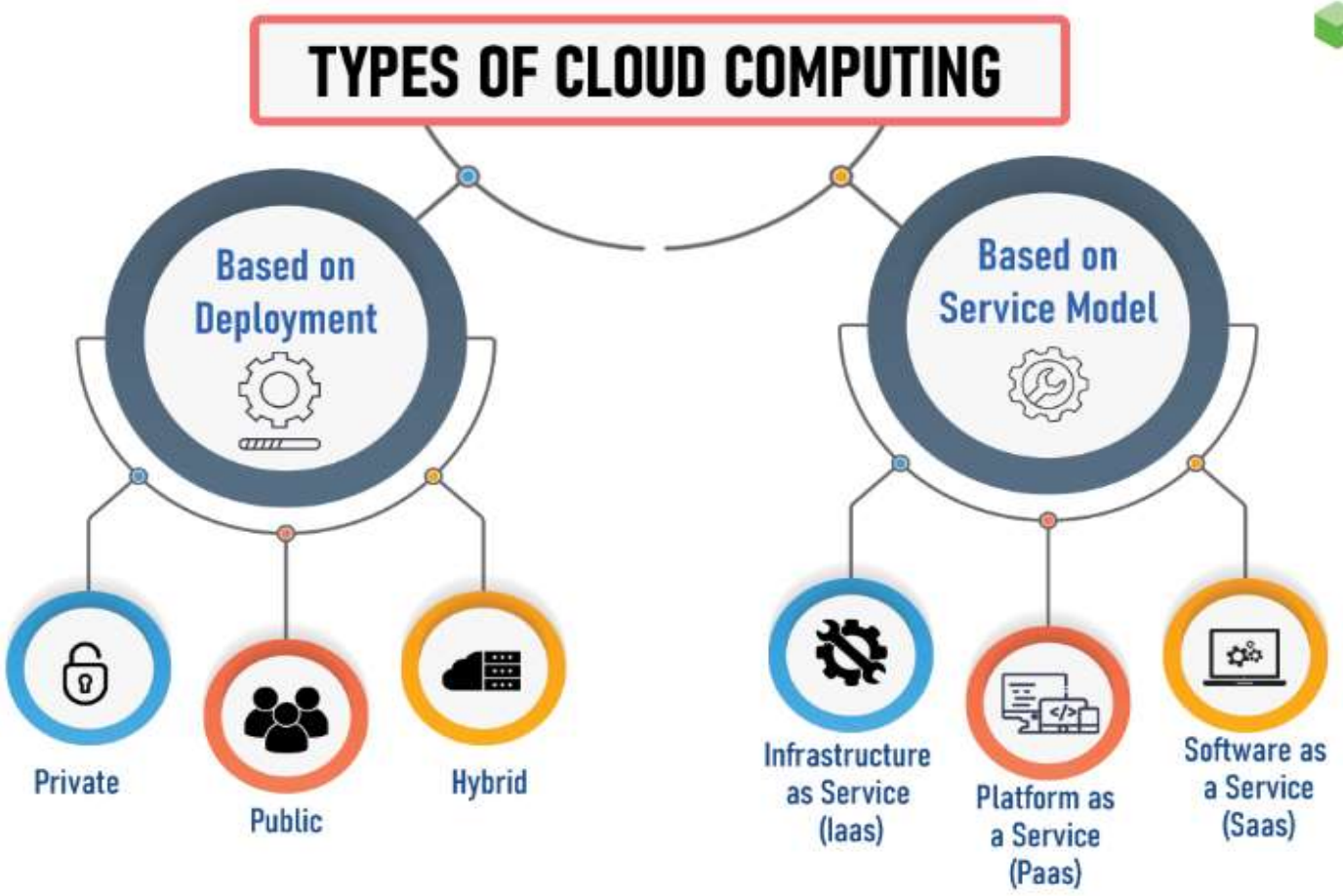


# Measured Service

* You get what you pay for.
* The resources usage can be monitored, controlled, and reported.
* Allows for accountability for the provider and consumer.



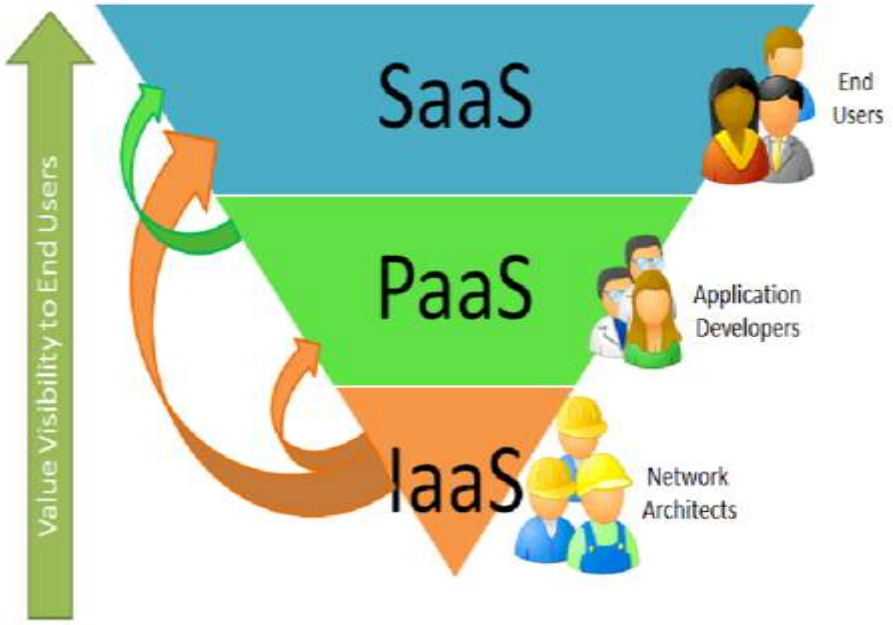
# Classification/types of Cloud Computing


TYPES OF CLOUD COMPUTING
Based on Deployment
Based on Service Model
Private
Public
Hybrid
Infrastructure as Service (Iaas)
Platform as a Service (Paas)
Software as a Service (Saas)

# Types/ classification based Service models

# Service models of cloud computing

1. **Software as a service (SaaS)**
2. **Platform as a Service (PaaS):**
3. **Infrastructure as a service (IaaS)**



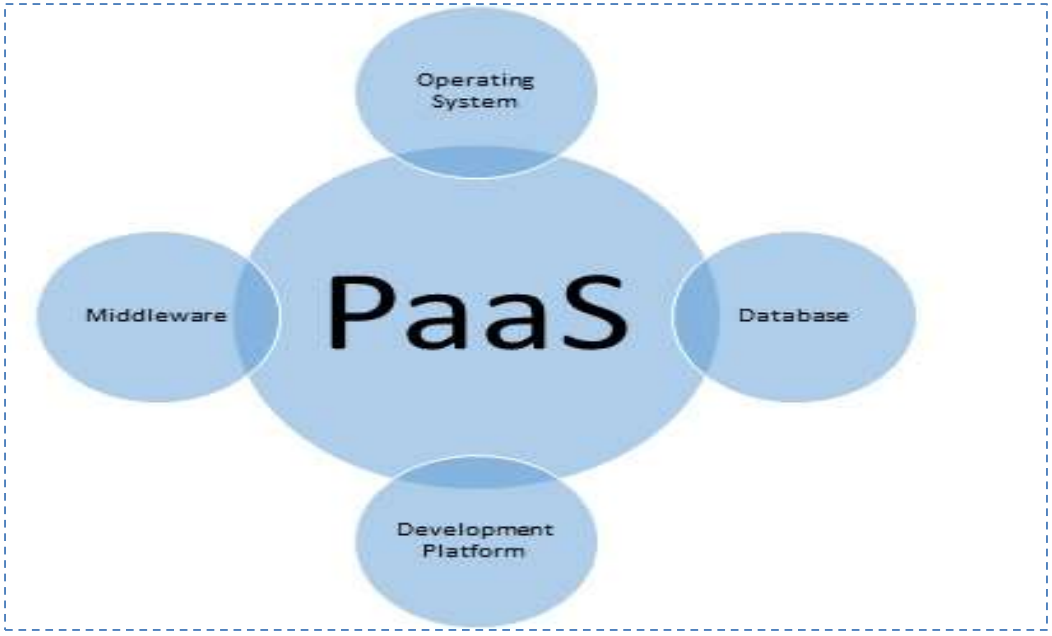
# Software as a service

- Provides consumer the capability to use applications running on a cloud infrastructure mainly on the web browser to access software that offer as a service over the web.
- The consumers do not have control the underlying framework including system, servers, network, operating systems, storage.
- Example for SaaS are Dropbox, Google Apps and GoToMeeting etc.



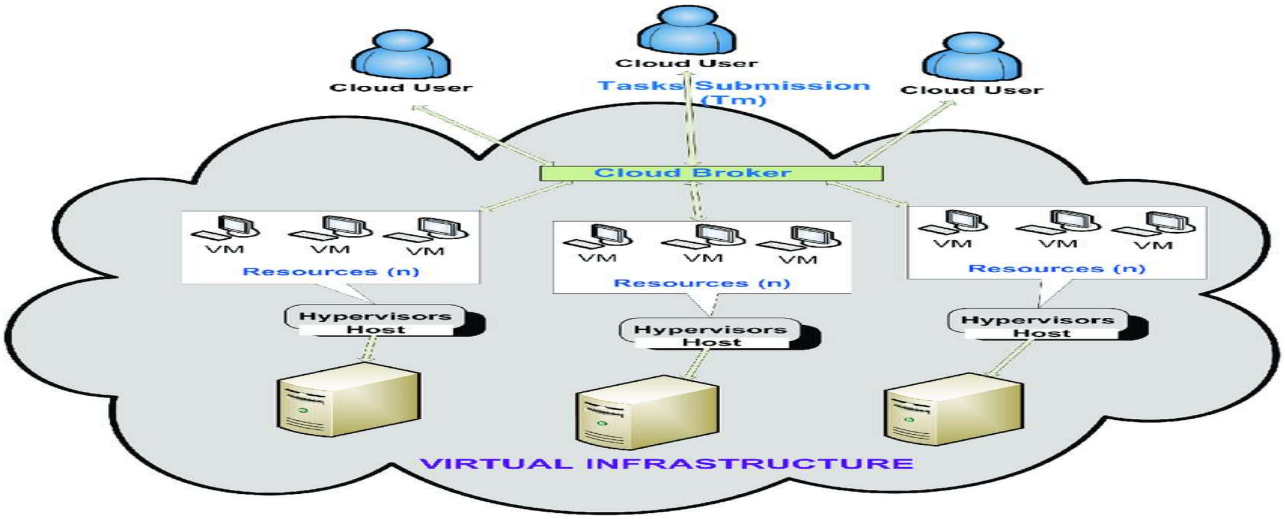
# Service model: PaaS

- Provides the capability to deploy development environment onto the cloud infrastructure.
- Consumer creates/develops applications using set of programming languages and tools that are supported by the provider.
- Examples of PaaS are **Google App Engine** and **Microsoft Azure** Phantom



# Service model: IaaS

- Provides the consumer with the capability to provision processing, storage, networks, and other fundamental computing resources and allow the consumer to deploy and run any software.
- The client has control over operating systems, storage, deployed applications and perhaps constrained control of select systems administration parts
- Examples of IaaS include *Amazon Web Services (AWS), Cisco Metapod, Microsoft Azure, Google Compute Engine (GCE*) etc.



# Service model: at a glance

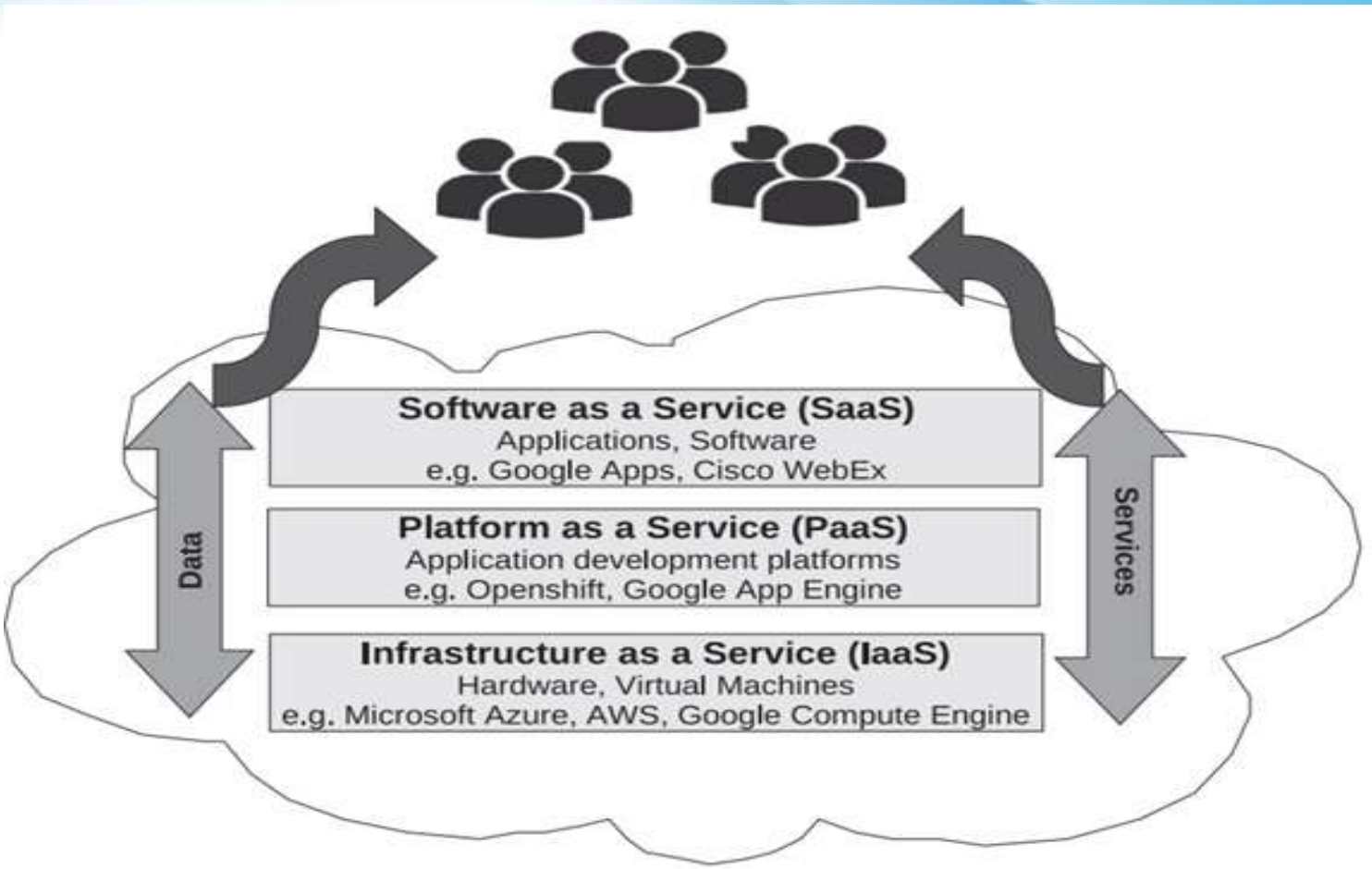


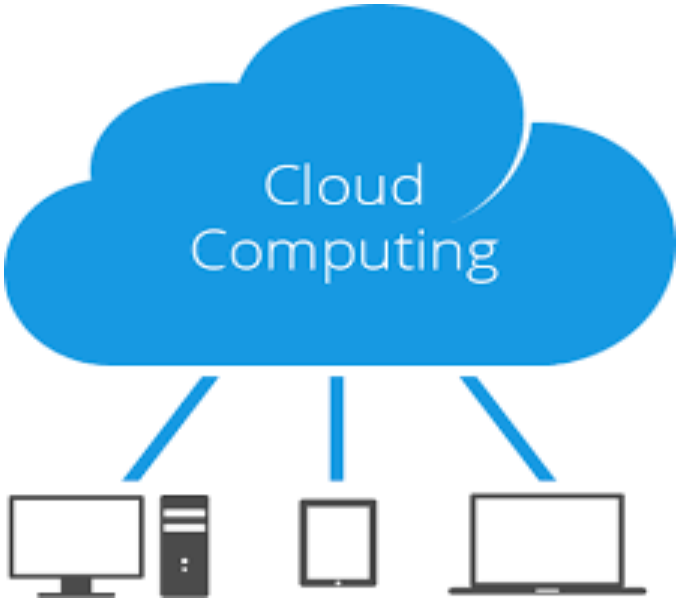

**Fig.** Cloud service delivery architecture.

# Classification/Types based on Deployment models

# *Deployment models of cloud computing*

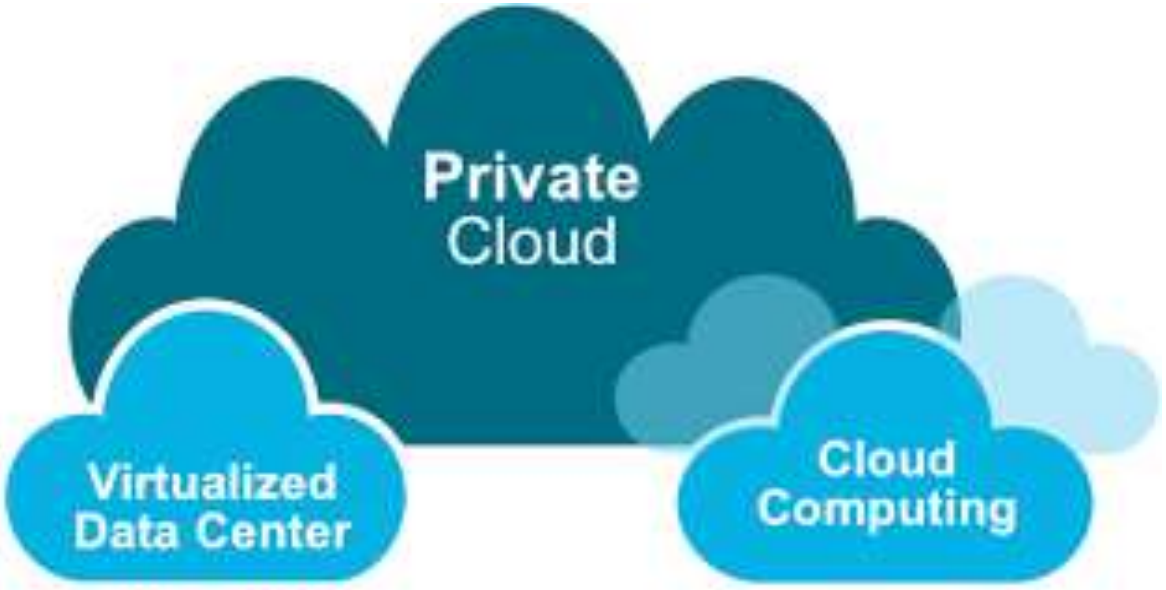
- A cloud organization model indicates how resources inside the cloud and shared.
- Deployment models are: **private cloud, public cloud, community cloud, and hybrid cloud**.

- Each model impacts the comparing scalability, reliability, security, and cost.



# *Private cloud*

- A cloud that is used exclusively by one organization, company, or one of its customers.
- Cloud is operated and maintained internally by a single organization or through a Third Party Auditing .



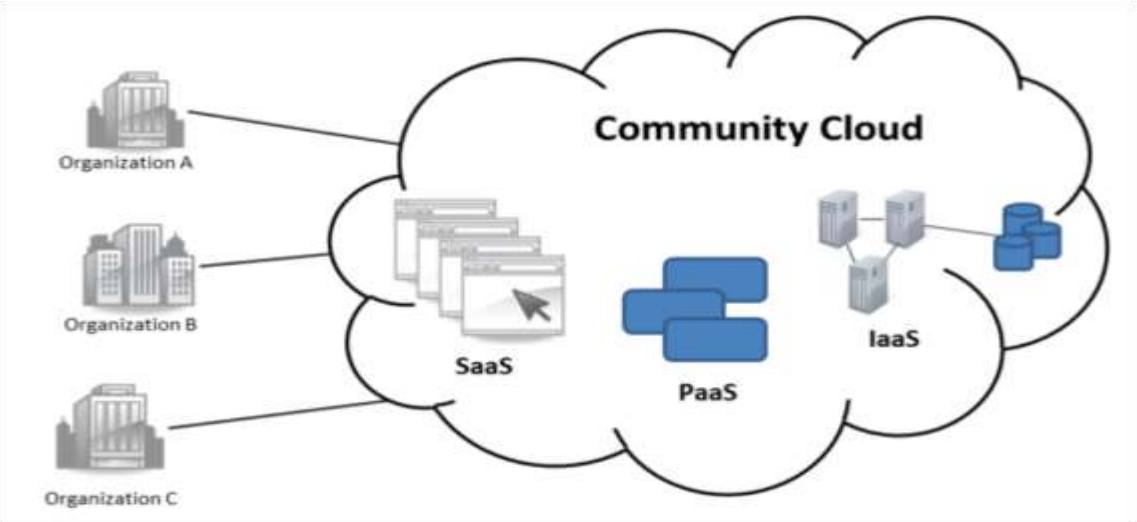
# Public cloud

- A public cloud is run and managed by the CSP and the physical infrastructure may presented at off-site location of the user.
- The cloud resources are shared among the multiple people and people pay to the cloud provider according to the services they use.



# Community Cloud

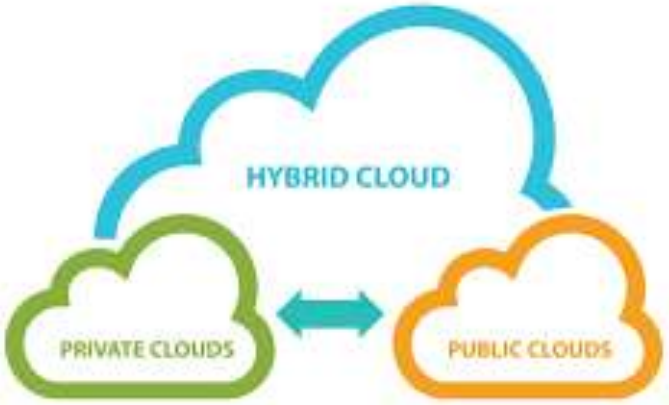
- Provides a cloud solution to a limited number of individuals or organizations that is governed, managed and secured commonly by all the participating organizations or a third party managed service provider.
- Often designed for businesses and organizations working on joint projects, applications, or research.



# *Hybrid Cloud*

- Uses a mix of on-premises, private cloud and third-party, public cloud services with orchestration between the two platforms.

- Allows workloads to move between private and public clouds as computing needs and costs change. Hybrid cloud gives businesses greater flexibility and more data deployment options.



# *Cloud role and boundaries*

The cloud provides different types of predefine role to the organization and human. These are:

- Cloud provider
- Cloud consumer
- Cloud service owner
- Cloud resource administrator
- Cloud broker
- Cloud carrier



# *Top leading Cloud Computing companies*

1. **Amazon Web Services(AWS)**

- One of the most successful cloud-based businesses is Amazon Web Services(AWS), which is an Infrastructure as a Service(Iaas) offering that pays rent for virtual computers on Amazon's infrastructure.

2. **Microsoft Azure Cloud Platform**

- Microsoft is creating the Azure platform which enables the .NET Framework Application to run over the internet as an alternative platform for Microsoft developers. This is the classic Platform as a Service(PaaS).

3. **Google Cloud Platform ( GCP )**

- Google has built a worldwide network of data centers to service its search engine. From this service, Google has captured the world's advertising revenue. By using that revenue, Google offers free software to users based on infrastructure. This is called Software as a Service (SaaS).

# Technologies in Cloud Computing

# Technologies in Cloud Computing

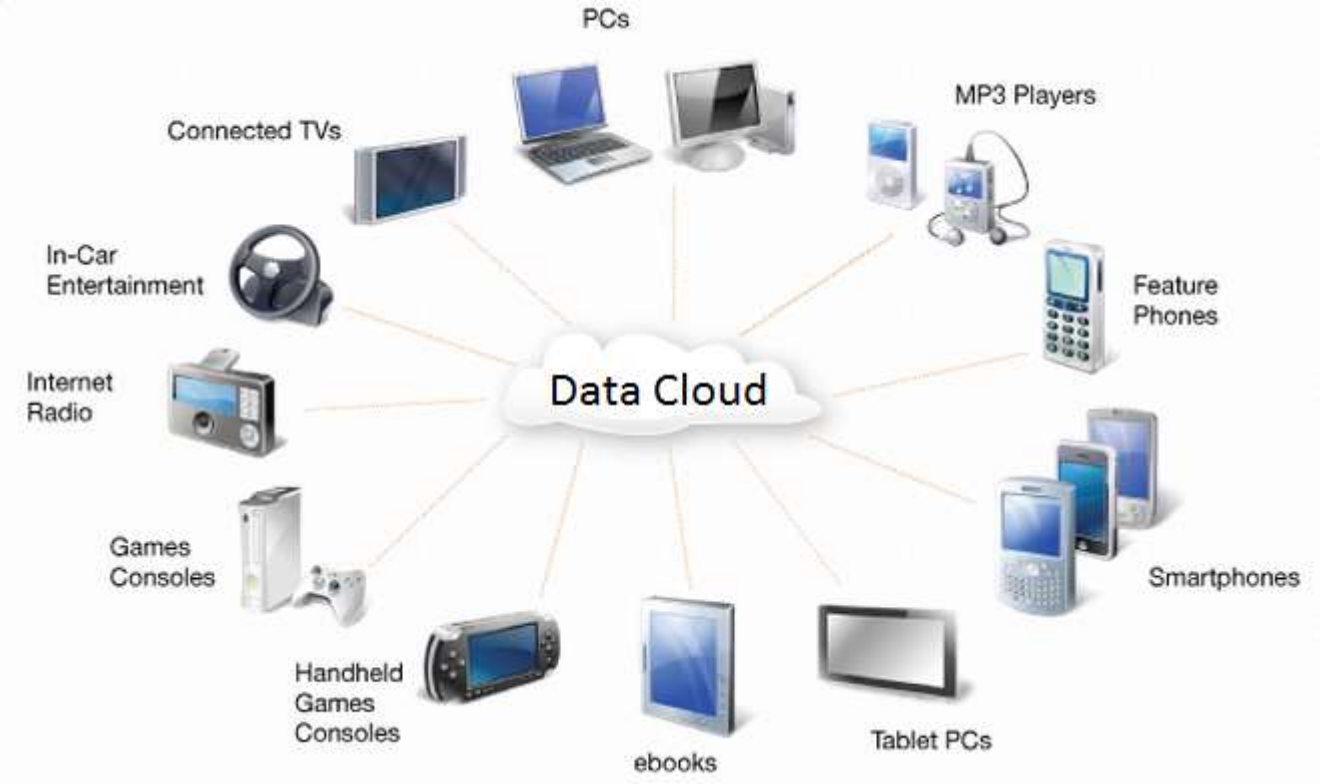
- Cloud computing isn't so much a technology as it is the combination of many pre-existing technologies.
- These technologies have matured at different rates and in different contexts, and were not designed as a coherent whole.
- These technologies have come together to create a technical ecosystem for cloud computing.

1. **Cloud access devices**
2. **Web technology**
3. **Broadband network and Internet technology**
4. **Data centers and server farms**
5. **Storage devices**
6. **Virtualization technologies**
7. **Service technology or API**
8. **Multi-tenant technology**



# Cloud access devices

- The range of access devices for the cloud has expanded in recent years like home & enterprise PCs, network computers, mobile phone devices, and custom static devices are all online.



# Web technology

-Users of multiple device types can now access applications and information from wherever they can load a browser.

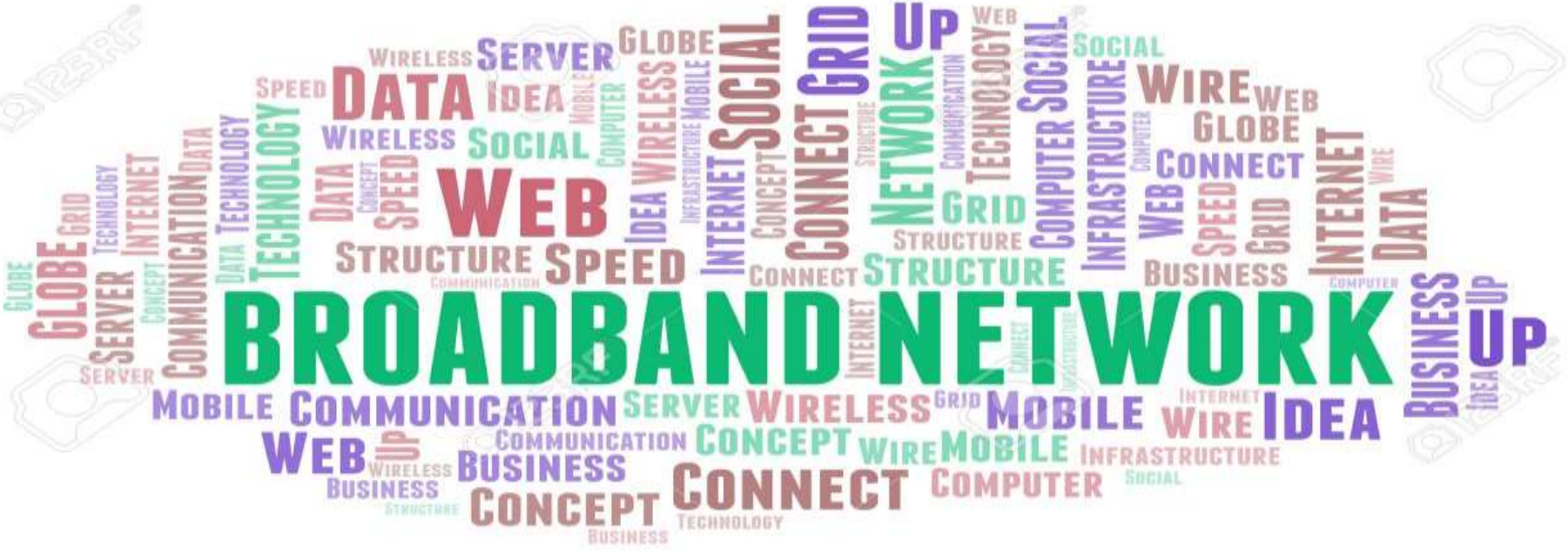
-The two fundamental constituent of the web technology is web client and web server. Proxies, routers, cache server, and gateways are known as the secondary components of the web



# *Broadband network and Internet technology*

- A critical component of the cloud is the broadband network, which offers the means to connect components.

-Broadband access is now widely available, especially in global metropolitan areas.



# *Data centers and server farms*

- Data center technology contains multiple technologies and components that are typically composed with one another.
- The data center has both physical and virtualized IT resources.



# Storage devices

-The fixed direct access storage device (DASD) has been replaced with storage area networks (SANs), which have reduced costs and allowed a great deal more flexibility in enterprise storage.

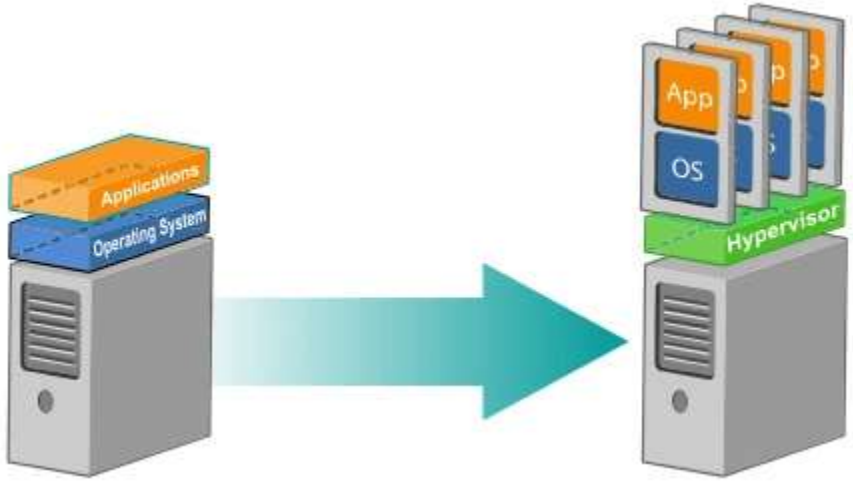
-SAN software manages integration of storage devices and can independently allocate storage space on demand across a number of devices.



# *Virtualization technologies*

-Virtualization refers to the abstraction of compute resources (CPU, storage, network, memory, application stack, and database) from applications and end users consuming the service.

-Virtualization technologies enable multi-tenancy cloud business models by providing a scalable, shared resource platform for all tenants.



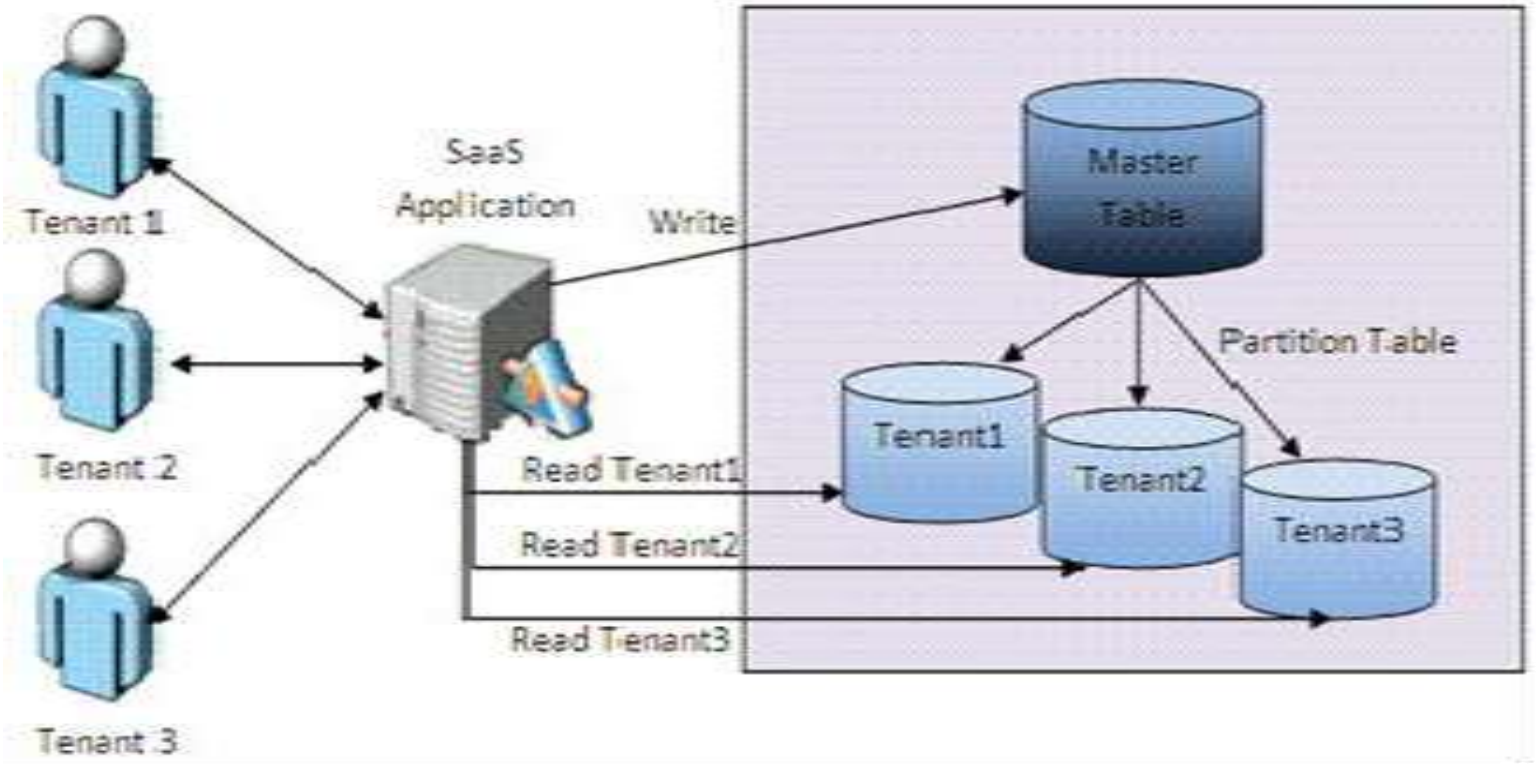
# Service technology or API

- Depending on the type of cloud services delivery model, an API can manifest in different forms, ranging from simple URL manipulations to advanced SOA-like programming models.
- APIs help to exploit the full potential of cloud computing and mask the complexity involved in extending existing IT management processes of the cloud services.



# Multi-tenant technology

- The same application logic is simultaneously accessed by multiple users using multi-tenant technology.
- The most common features of multi-tenants application is usage isolation, data recovery, data tier isolation, data security, application upgrades, system scalability, and metered usage



# Traditional IT vs Cloud Computing

TABLE. *Cloud computing: A customer's perspective*

| Dedicated/traditional IT | Cloud computing |
|---|---|
| High upfront IT investments for new builds | Low upfront IT investments; pay-for-use model |
| High cost of reliable infrastructure | Reliability built into the cloud architecture |
| High complexity of IT environment | Modular IT architecture environments |
| Complex infrastructure | No infrastructure |

# Key drivers of adopting cloud

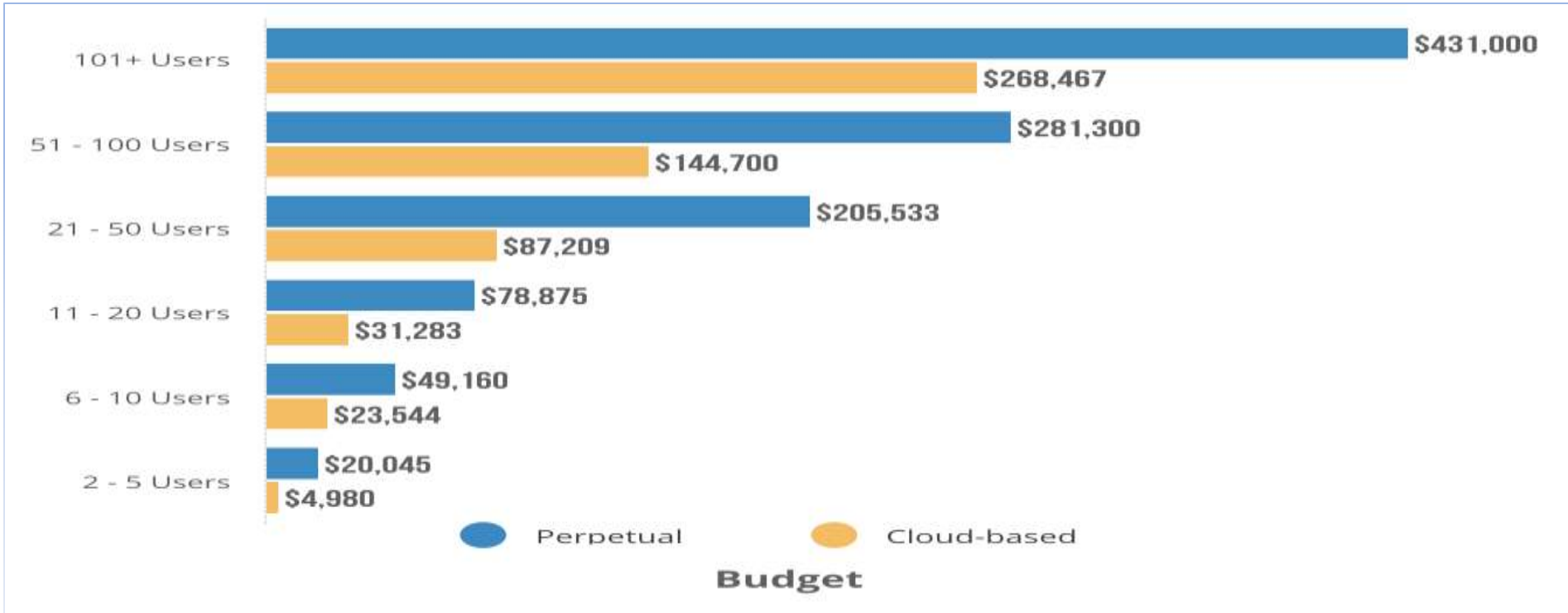
# *Key drivers of adopting cloud*

**There are a number of compelling reasons to move operations toward cloud computing.**

- **Small Initial Investment and Low Ongoing Costs**
- **Economies of Scale**
- **Open Standards**
- **Sustainability**

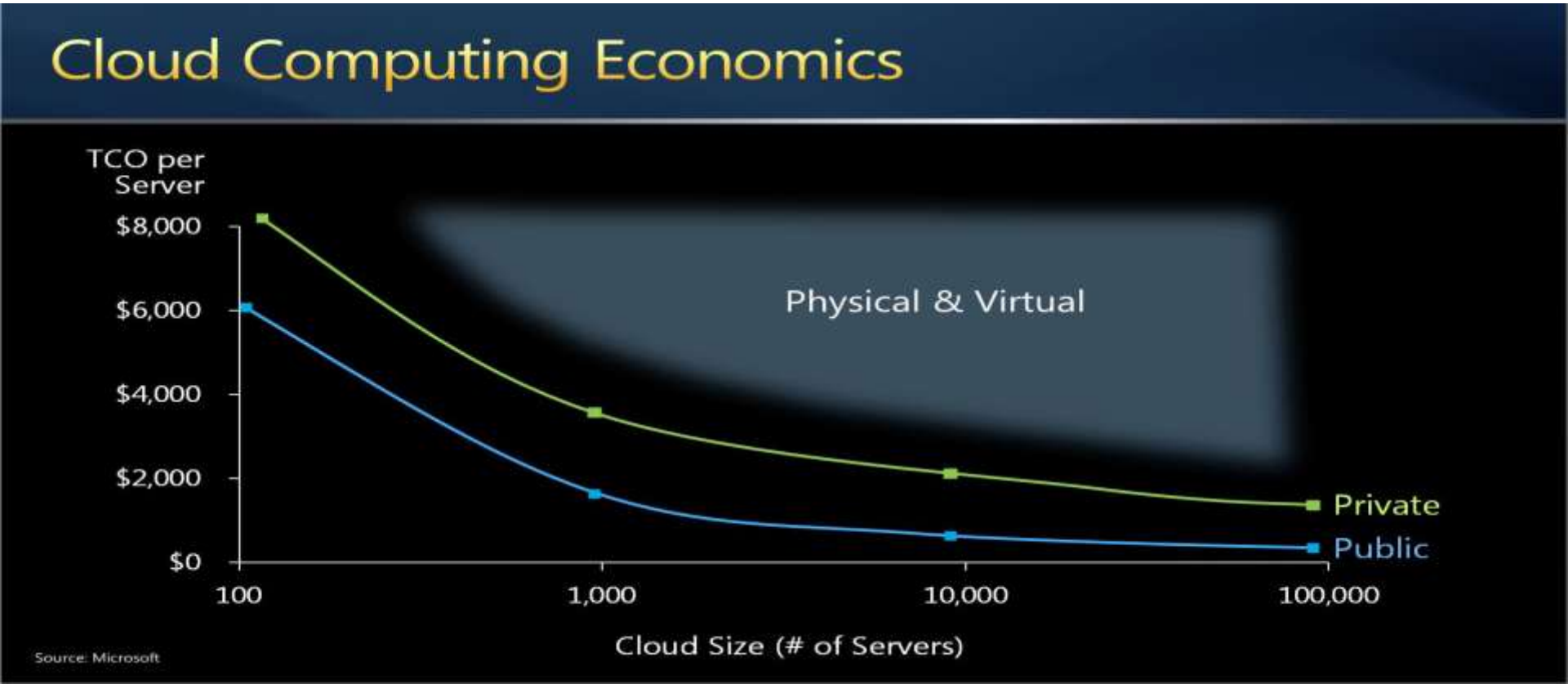
# *Key Drivers: Small Initial Investment and Low Ongoing Costs*

- Public cloud computing can avoid capital expenditures because no hardware, software, or network devices need to be purchased.
- Cloud usage is billed on actual use only, and is therefore treated  no more as an expense venture.



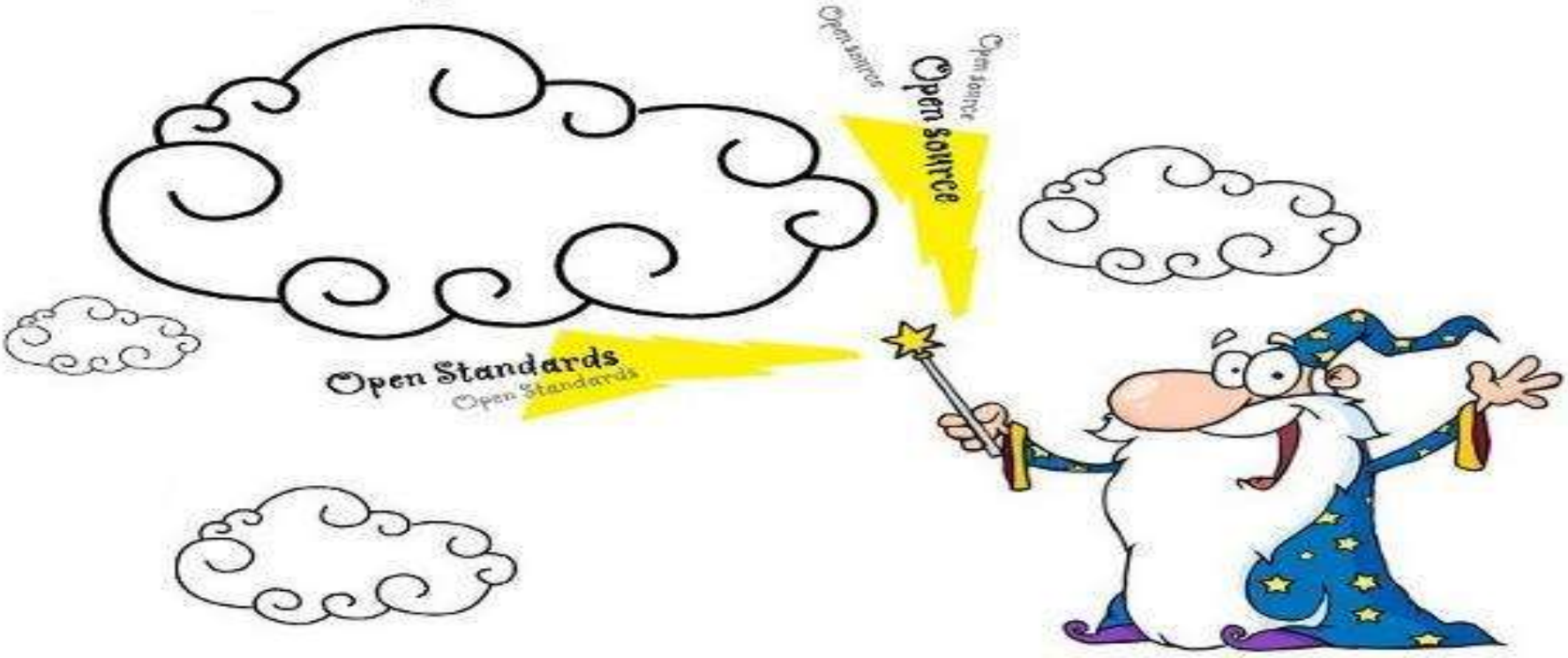
# Key Drivers: Economies of Scale

- Most development projects have a sizing phase during which attempts to calculate the storage, processing power, and memory requirements during development, testing, and production.
- With the flexibility that cloud computing solutions offer, companies can acquire computing and development services as needed and on demand,



# Key Drivers: Open Standards

- Some capabilities in cloud computing are based on open standards for building a modular architecture that can grow rapidly and can change when required.
- The flexibility to alter the source code is essential to allow for continued growth in the cloud solution.



# *Key Drivers: Sustainability*

- CSPs have invested considerable expense and thought into creating a resilient architecture that can provide a highly stable environment
- Cloud computing allows companies to rely on the CSP to have limited points of failure, better resilience via clustering, and the ability to invest in state-of-the-art resilience solutions



# *Uses of cloud computing*

From tiny start-ups and global corporations to government agencies and nonprofits — are embracing the technology for a variety of reasons and use cases.

- **Create new apps and services**: Quickly build, deploy, and scale applications like web, mobile, and API on any platform. Access the resources you need to help meet performance, security, and compliance requirements. Build, test and take down environments quickly, speeding time to market for new developments.

- **Store, backup, and recover data**: Avoid outages and protect your data cost-effectively — and at massive scale — by transferring your data over the internet to an offsite cloud storage system that's accessible from any location and any device. Improve disaster recovery efforts to ensure business continuity.

- **Stream audio and video**: Connect with your audience anywhere, anytime, on any device, with high-definition video and audio, with global distribution.

- **Deliver software on demand**: Also known as software as a service (SaaS), on-demand software lets you offer the latest software versions and updates to your customers — anytime they need them, anywhere they are.

- **Work with big data analytics and AI**: Use the powerful processing capabilities of the cloud to collect and analyze big data or to power artificial intelligence and machine learning applications.
- **Support your mobile workforce**: Give your mobile and remote workers virtual desktops they can easily access from any laptop, tablet, or mobile device. Keep workers connected with cloud-based email and messaging platforms.

# Edge computing

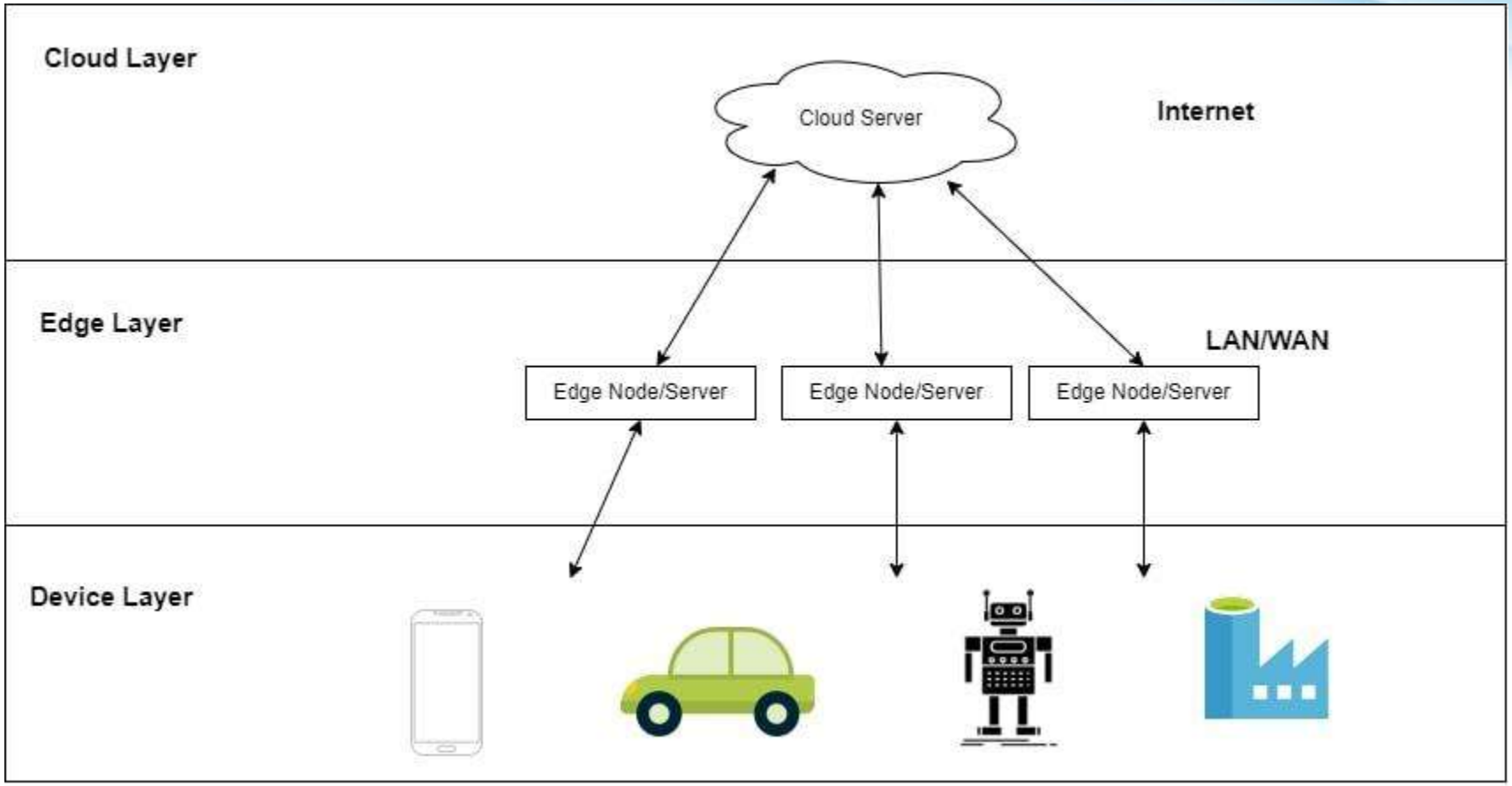
- Edge computing is a subsection of cloud computing.

- While cloud computing is about hosting applications in a core data centre, edge computing is about hosting applications closer to end users, either in smaller edge data centres or on the customer premises instead.

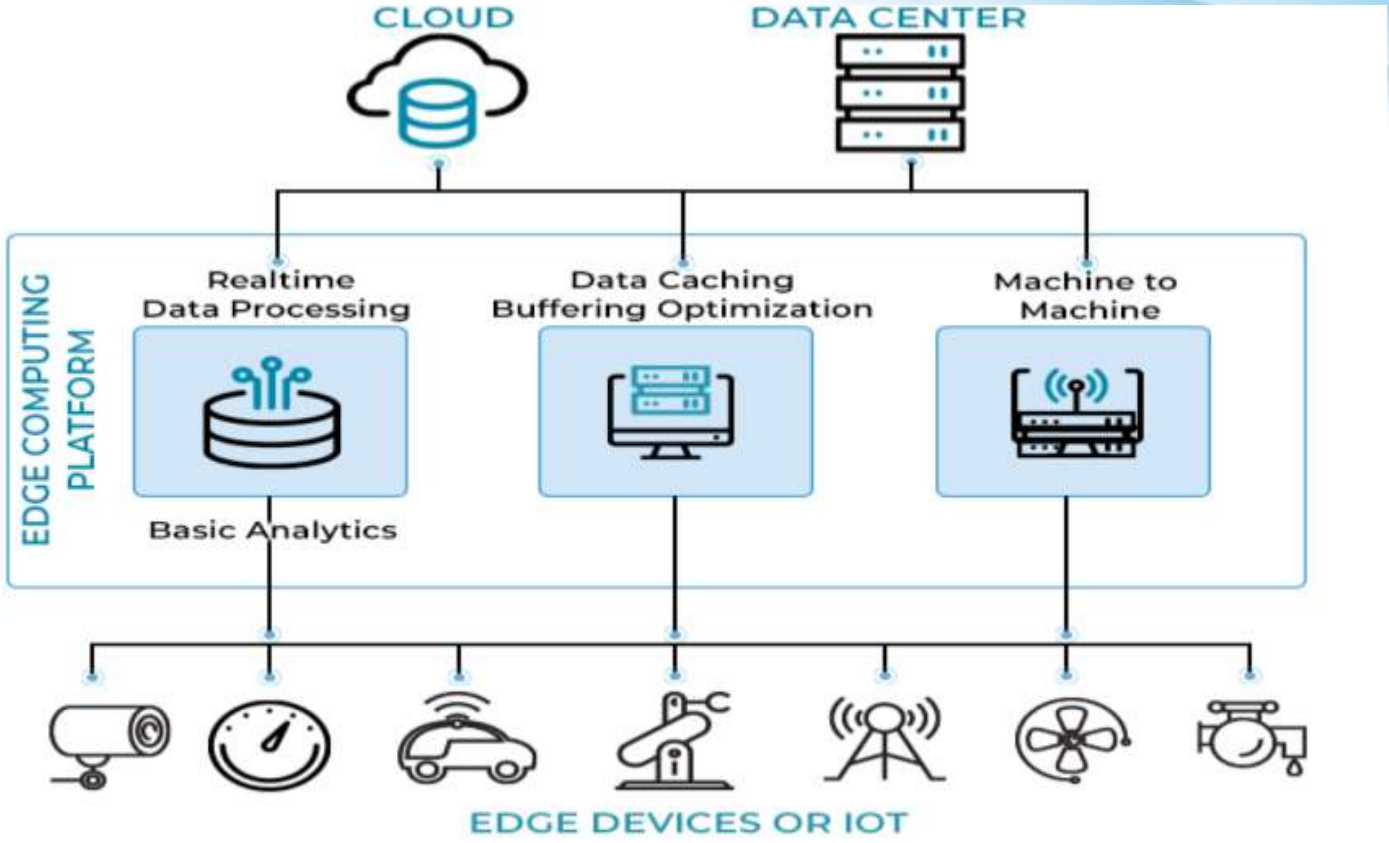
# Edge computing

- Edge computing optimizes Internet devices and web applications by bringing computing closer to the source of the data.
- This minimizes the need for long distance communications between client and server, which reduces latency and bandwidth usage.
- Edge computing means running fewer processes in the cloud and moving those processes to local places, such as on a user's computer, an IoT device, or an edge server. Bringing computation to the network's edge minimizes the amount of long-distance communication that has to happen between a client and server.

# Edge computing



# Edge computing




Edge computing

# Example of edge computing

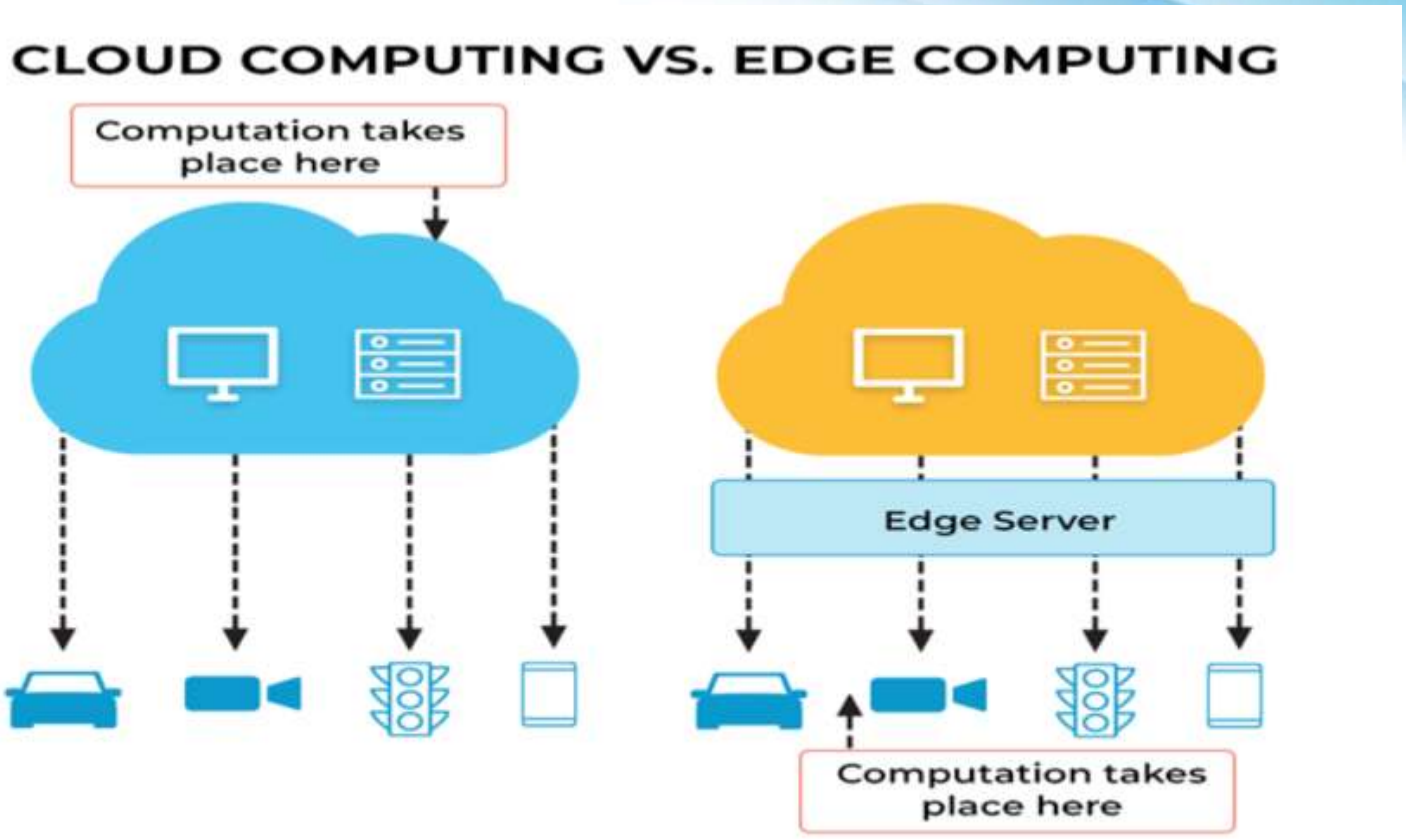
- Consider a building secured with dozens of high-definition IoT video cameras. These are "dumb" cameras that simply output a raw video signal and continuously stream that signal to a cloud server.
- On the cloud server, the video output from all the cameras is put through a motion-detection application to ensure that only clips featuring activity are saved to the server's database.
- This means there is a constant and significant strain on the building's Internet infrastructure, as significant bandwidth gets consumed by the high volume of video footage being transferred.
- Additionally, there is very heavy load on the cloud server that has to process the video footage from all the cameras simultaneously.

# Example of edge computing

- Now imagine that the motion sensor computation is moved to the network edge. What if each camera used its own internal computer to run the motion-detecting application and then sent footage to the cloud server as needed? This would result in a significant reduction in bandwidth use, because much of the camera footage will never have to travel to the cloud server.

- Additionally, the cloud server would now only be responsible for storing the important footage, meaning that the server could communicate with a higher number of cameras without getting overloaded. This is what edge computing looks like.

# Differences between cloud and edge computing




Difference Between Cloud Computing and Edge Computing

# *Differences between cloud and edge computing*

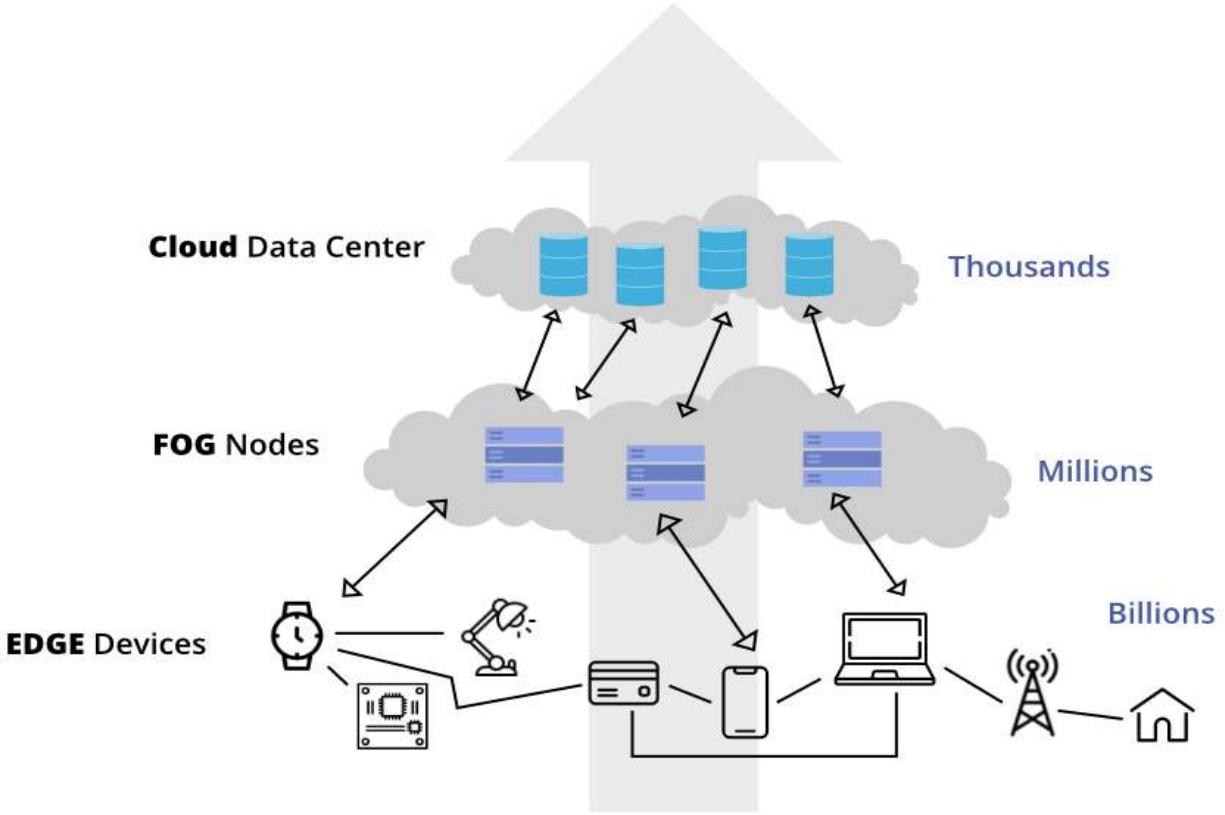
| Computing | Edge Computing |
|---|---|
| **Cloud computing is centralised servers stored in faraway, large-scale data centres.** | On the other hand, it is a highly distributed and global computing infrastructure closer to the devices and users. |
| **It processes data on a central cloud server far from the data sources.** | Process data on site quickly and analyze data in real-time. It does not focus on storing data. |
| **It is suited for applications that are not time-sensitive.** | It is ideal for low latency, where every millisecond counts. |
| **Cloud computing gives improved and innovative processing capabilities and storage capacity.** | As it processes in the device, it has lower processing power and storage capacity. |
| **It is suitable for in-depth and long-term analysis.** | It is better for fast and real-time analysis. |
| **It needs internet connectivity.** | It can work without internet connectivity. |
| **An expensive and intensive operational activity for the company.** | Automated scalability with zero-touch provisioning |
| **The connectivity, data movement, bandwidth, and latency cost are relatively high.** | Fewer latency and bandwidth requirements increased performance and lower operational expenses. |

# Fog Computing

Fog computing, also known as fog networking, is a distributed computing architecture that brings cloud computing capabilities to the edge of a network. It allows devices to perform computation, storage, and communication locally, and then route the data over the internet



# Fog Computing




Fog computing is a layer in between the edge and the cloud. When edge computers send huge amounts of data to the cloud, fog nodes receive the data and analyze what's important. Then the fog nodes transfer the important data to the cloud to be stored and delete the unimportant data or keep them with themselves for further analysis. In this way, fog computing saves a lot of space in the cloud and transfers important data quickly

# Thank You

